

DILIP BOIDYA

Likabali Alikash, Dhemaji, Assam, India

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Technical Skills

Core Skills: HTML5, CSS3, JavaScript, ReactJS, NodeJS, MongoDB, PostgreSQL, Flutter

Tools: Git, GitHub, Docker, Postman, VS Code, Chrome DevTools

Hardware Skills: Circuit design and PCB design, Microcontrollers(ESP12F, Arduino), Sensors, Soldering.

Tools: KiCad, CST Microwave suite.

Education

Tezpur University

Aug 2018 – Jun 2022

B.Tech in Electronics and Communication Engineering

Tezpur, Assam

Experience

Freelancer

October 2025 – Present

Freelancing

Remote

- **VitalsSense Module:** Assisted in PCB fabrication and assembly for a vitals monitoring module using MLX90614, MAX30100, and Sreed Studio XIAO Sense (MG24 MCU).
- **IoT – Raspberry Pi HAT PCB:** Designed a Raspberry Pi HAT with six relay channels and six digital input interfaces for automation and control systems.
- **IoT – ESP32 Controller (Nirvana Technology):** Designed an ESP32-WROOM-32D-based IoT controller integrating ten relay output channels.

Intern

June 2024 – September 2024

Akrabhi Software Pvt. Ltd.

Remote

- Developed a full-stack productivity tool for bookmark management.
- Designed and implemented RESTful APIs and worked with databases for backend functionality.

Open Source Contributor

January 2024 – February 2024

Chingu

Remote

- Contributed to a task management application enabling users to plan, manage, and complete tasks efficiently.
- Live Demo — GitHub Repository

Fellow

May 2022 – July 2022

FOSSEE, IIT Bombay

Remote

- Modeled and simulated semiconductor devices including diodes, LEDs, and transmission lines for an open-source EDA tool.

Projects

Pure Air — Real-Time Air-Quality IoT Platform

Devpost Hackathon 2025

- Designed and implemented an MQTT-based IoT platform using ESP12F (simulated) and HiveMQ Cloud.
- Built a Node.js backend for real-time data ingestion and control signals.
- Developed a React Native app for AQI visualization, remote purifier control, and live alerts.
- **Links:** Devpost — GitHub (App) — Deployed App (APK)

QnA Platform for Learners

- Developed MERN stack QnA platform for learners and SMEs with separate frontends and REST APIs.
- Learner App — SME App

B.Tech Final Year Project — Directional Coupler

- Designed, simulated, and fabricated a Microstrip Line Directional Coupler; awarded 3rd prize (Prof. Bipin Dhekial Phukan Memorial Award).
- Project Blog

Achievements

Cloud-based Analog IC Design Hackathon

2023

- Designed an LNA in Synopsys Custom Compiler; achieved **Excellent** rating.
- GitHub

Mixed Signal Circuit Design Marathon

2023

- Designed a Step-up Converter using eSim + Verilog; achieved **Excellent** category rating.
- GitHub

Circuit Design & Simulation Marathon

2022

- Designed a Buck Converter using eSim tool (Sky130nm tech); achieved **Very Good** category rating.
- GitHub